technology can be harnessed and directed in ways that minimise disruption of the environment.[19]

While the explicit consideration of environmental objectives and constraints in product and process development can lead to ‘green’ technology solutions, in reality, the question of what is green depends on how environmental problems are defined. As problem definitions change, solutions also change. Basically, one’s philosophical view of the relationship between human activity and the environment strongly conditions one’s understanding of the role of technology and the ‘environmental problems’ that technology can potentially address. For better understanding of the issue and the possible wayouts we may discuss the ***three alternative ideas*** that highlight the different roles that technology can play with respect to the environment.[20]

Environmental Protection

In this idea, the environment is recognised as an *economic externality*[21] that must be safeguarded through laws and regulations. Tradeoffs are seen between industrial competitiveness and protecting the environment – such as employment vs. protecting endangered species – and cost-benefit analysis is offered as a means of balancing the two. This view is fundamentally anthropocentric (which means humans as more important than anything else), with the principal concern being the effect of pollution on human health and welfare.

In this case, the ‘problem’ is that human society produces too much waste. This is why this idea leads to policies that focus on reducing the quantity or toxicity of waste, e.g., waste prevention, recycling, or waste treatment. Consequently, technology should be used to reduce the quantity and toxicity of wastes requiring disposal, for example, making products more recyclable, light-weighting, eliminating hazardous materials, etc. Progress is measured in terms of increasing efficiency of energy and materials use; that is, reducing the quantity of energy and materials required per unit of production. This view does not concern itself explicitly with whether the physical flows of energy and materials through the economy are ‘ecologically sustainable’.